Ref. [27] — Sharp Minima Can Generalize for Deep Nets

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1. The research question: does flatness explain generalization?

Keskar et al. connected large-batch training with sharp minima and a test-performance gap. Dinh et al. ask a more fundamental question: *is ordinary parameter-space sharpness a property of a trained function, or merely of how that function is parameterized?* If two parameter vectors give identical predictions for every input, they have identical training and test performance; a measure intended to explain generalization should ideally not give conflicting conclusions for them.

2. Method: exploit the exact symmetry of ReLU networks

The key identity of rectified linear units is *positive homogeneity*:

$$\sigma(cx) = c\sigma(x) \qquad (c > 0), \quad \sigma(z) = \max\{0, z\}.$$

For a simple two-layer model, $f(x; W_1, W_2) = W_2\sigma(W_1 x)$, choose any $c > 0$ and change weights to

$$W_1' = cW_1, \qquad W_2' = c^{-1}W_2.$$

Then, for every x,

$$f(x; W_1', W_2') = c^{-1}W_2\sigma(cW_1 x) = W_2\sigma(W_1 x) = f(x; W_1, W_2).$$

Hence a loss that depends only on predictions obeys $L(W_1', W_2') = L(W_1, W_2)$, even though the coordinates in weight space can change by an arbitrarily large scale factor. The same symmetry extends to appropriate units and layers of rectifier networks.

3. Why this changes sharpness: first-principles calculation

A one-dimensional *teaching model*, not a theorem quoted from the paper, makes the coordinate issue transparent. Let $L(a, b) = (ab - 1)^2$ and take the global minimum $(a, b) = (c, c^{-1})$. Its Hessian is

$$\nabla^2 L(c, c^{-1}) = 2\begin{pmatrix} c^{-2} & 1 \\ 1 & c^2 \end{pmatrix}.$$

The eigenvalues are 0 and $2(c^2 + c^{-2})$. Thus, at equally good global minima representing the *same* product $ab = 1$, the largest Hessian eigenvalue grows without bound as $c \to \infty$. Curvature is not invariant under this symmetry. Dinh et al. establish analogous problems for definitions of flatness and sharpness in actual rectified networks.

Main result. Reparameterization/symmetry can make functionally equivalent deep-network solutions arbitrarily sharper by common parameter-space measures, while predictions and generalization remain unchanged. Therefore naive sharpness alone cannot be a universal explanation for generalization.

4. Assumptions and limits

The central construction relies on parameter symmetries, particularly positive homogeneity of rectifier architectures and appropriate losses. It does *not* prove that all flatness measures are useless, that sharpness never correlates with generalization empirically, or that large-batch generalization gaps are imaginary. Geometry-aware or reparameterization-invariant notions may still be informative.

Survey relevance (RQ1, RQ4). This is a crucial qualification to "flat minima generalize": compare geometric measurements across models only after stating the parameterization and normalization. The conclusion is a counterexample to universal interpretation, not a claim that SGD does not prefer particular basins.

5. Our notes / comparison with Ref. [26]

Keskar et al. report an empirical association between batch size, sharpness and generalization. Dinh et al. show why the plain weight-space sharpness in that explanation can change without changing the model's function. The two papers are best read together: empirical correlation is not a coordinate-invariant causal explanation. **Original source:** https://proceedings.mlr.press/v70/dinh17b.html (main paper and supplement).